

Predation Risk as a Cost of Reproduction

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Predation risk as a cost of reproduction in animals has recently received increased empirical and theoretical attention. Higher risk may be associated with all stages of reproduction. Examples of evolutionary responses to this increased risk include habitat choice, duration and timing of display and copulation, changes in brightness of breeding coloration, and changes in life history traits such as age of reproduction and reproductive effort.

The cost of reproduction is defined as a trade-off between present and future reproduction. Reproductive investments made now can cause a decrease in future reproductive success. This can be due, on the one hand, to energy or nutrient exhaustion leading to a negative correlation between present and future fecundity – the 'fecundity cost'¹. On the other hand, reproductive investment can lead to a decreased probability of an individual's own survival, caused either by metabolic drain or by an increased susceptibility to pathogens and predators – the 'survival cost'¹. Evidence that a survival cost is imposed by the riskiness of reproductive activities is scarce but the number of studies on predation risk as a reproductive cost is currently increasing.

In a recent *TREE* commentary, Gwynne² discussed the hypothesis that copulating animals are at greater risk of being eaten. However, an increased predation risk caused by reproductive activities can arise during all stages of reproduction. Behaviours performed in order to acquire a partner may involve an increased mobility due to searching activities^{3,4} (and hence increased probability of predator encounters) and/or a conspicuous display that may attract predators as well as mates^{5,6}. Also, egg carrying may increase an animal's conspicuousness^{7,8}, and a reduced agility can make gravid females easier to catch than other individuals⁹.

There is often a conflict between maximizing the probability of finding a mate and minimizing the risk of being detected by a predator, since conspicuousness may be crucial in the former but deadly in the latter.

Thus, there are contrasting selective pressures from sexual selection, which is a result of differential mating success, and natural selection, arising from variance in all other components of fitness, for example survival¹⁰.

Risks of mate-searching activities

In males, reproductive success often depends on the number of mates fertilized, while females in general are restricted by the number of eggs produced [with the exception of sex-role-reversed animals such as pipefish (*Syngnathidae*)¹¹]. It follows that males are usually the sex most active in mate-acquisition behaviour, and consequently most likely to encounter predators. For example, male isopods (*Idothea baltica*) were more active during the day than females in the breeding season, and it was suggested that this would lead to higher predation on males¹². This could account for the observed female-biased sex ratio towards the end of the breeding season. In the tick-tock cicada (*Cicadetta quadricincta*), males made many short flights in order to change the perch from which to sing or to move towards receptive females, and were captured by web-building spiders more often than were females³.

For short-lived species, it is important for males to locate females as quickly as possible. This sometimes leads to indiscriminate mate-location behaviour⁴, resulting in misdirected courtship. If the cost of making mistakes is low, this behaviour is probably advantageous, but Moore (see Ref. 4) showed that such indiscriminate mate location actually may increase male mortality in the butterfly *Euphydryas editha*. These male butterflies approach and attempt copulations with many objects resembling females, such as males and dead individuals. More males than females are caught in spider webs, and this is partly due to their higher activity. However, when webs already containing butterflies were compared with webs from which old butterflies had been removed,

the increase in number of male *E. editha* was significantly higher in the former, indicating that the misdirected attraction to dead conspecifics increased male mortality rate⁴.

Signalling for mates (... and predators)

An alternative to active mate searching is to be stationary and attract mates by some kind of display. This display can consist either of visual cues such as a breeding coloration and/or a courtship behaviour, as commonly occurs in fish¹³, or of acoustic cues such as the calling in frogs⁶, crickets¹⁴ and cicadas³. Furthermore, olfactory cues can be used by both females¹⁵ and males¹⁶ in order to attract mates.

Such signals may increase conspicuousness to predators (reviewed in Ref. 5) (Table 1). For animals with an acoustic display it is not always the calling male that suffers the increased mortality. In frogs, males are located by their calls and caught by various predators such as bats, toads, opossums and crabs⁶. In contrast, in the cricket *Gryllodes supplicans*, the male calls from a burrow, where he is protected from predators; geckos orient to male cricket calls but actually consume the female crickets that are also attracted by the calls¹⁴.

Risks of copulation

Increased risks of being eaten during copulation include cases where the male is eaten by the female, as in mantids¹⁷; but an increase in predation risk from other species during mating (including both actual copulation and amplexus) has also been observed^{18,19}. In amphipods (*Gammarus* spp.), pairs in amplexus are larger and slower-moving prey units than single animals and are heavily preyed upon by fish in some circumstances¹⁸. Similarly, individual water striders (*Gerris remigis*) can escape fish predators faster than can mating animals¹⁹. However, as discussed by Gwynne², in some cases, such as in male milkweed beetles (*Tetraopes tetraophthalmus*)²⁰, mating actually

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Table 1. Studies in which it has been shown that signalling for mates also attracts predators or parasites

Signal	Prey	Sex ^a	Predator	Ref.
Acoustic	Frog, <i>Physalaemus pustulosus</i>	♂	Bat, <i>Trachops cirrhosus</i>	6
		♂	Opossum, <i>Philander opossum</i>	6
		♂	Bull frog, <i>Leptodactylus pentadactylus</i>	6
		♂	Crab, <i>Potamosarcinus richmondi</i>	6
	Cricket, <i>Gryllodes integer</i>	♀	Gecko, <i>Hemidactylus tursicus</i>	14
Visual	Guppy, <i>Poecilia reticulata</i>	♂	Various piscivores, e.g. Cichlidae, Cyprinodontidae	30
	Firefly, <i>Photinus collustrans</i>	♀	Spiders, Lycosidae	32
	Fireflies, e.g. <i>Photinus</i> spp.	♂	Fireflies (♀), <i>Photuris</i> spp.	33
Olfactory	Mouse, <i>Peromyscus maniculatus bairdi</i>	♀	Weasel, <i>Mustela nivalis</i>	15
	Stink bug, <i>Nezara viridula</i>	♂	Tachinid fly, <i>Trichopoda pennipes</i>	16

^a The sex with an increased vulnerability to predators during reproduction is indicated.

reduces predation due to reduced locomotor activity when paired. Thus, the effects of decreased agility can be either negative (through decreased escape capacity) or positive (through decreased conspicuousness).

Pregnancy, egg carrying and incubation

Gravid females often have lower agility than other individuals^{9,21}. A decreased locomotor ability during gestation was found in the garter snake (*Thamnophis marcianus*)⁹; in the grass snake (*Natrix natrix*), gravid females were found to have a lower survival rate than other individuals, probably because of higher predation by birds²². However, female lizards of the species *Eumeces laticeps* avoid an increased predation risk by changing their behaviour while gravid²¹: usually, they run to escape predators, but when

gravid they decrease their activity and rely instead on crypsis.

Thus, animals carrying eggs can experience an increased risk of predation, as a result of both higher conspicuousness and decreased mobility. Reproducing female prawns (*Palaemon adspersus*) were eaten by fish at a higher rate than were non-reproducing females⁷. In the pipefish *Nerophis ophidion*, males brood the eggs on their bodies and are then more susceptible to predation than otherwise⁸.

Most studies on birds have concerned nest predation, i.e. predation on eggs and nestlings. Only a few have shown that reproduction can increase the risk of predation on parents; for instance, in the scarlet rosefinch (*Carpodacus erythrinus*)²³ and in the pied flycatcher (*Ficedula hypoleuca*)²⁴, females were found to be taken by predators while incubating.

Adaptations to predation risk

There are many examples of morphological and behavioural adaptations for predator evasion in general. However, studies of evolutionary responses to an increased predation risk during reproduction are rare. Exceptions are the many studies on the guppy *Poecilia reticulata*, dealing with selection, life history and reproductive behaviour²⁵⁻³¹ (Table 2).

It was discovered early on that where predation risk is high, male guppies decrease the rate of courtship display²⁵ and also become less

colourful²⁶. Recently, an array of other responses to predation risk has been discovered in this species. Where predators are abundant, females show a lower preference for brightly coloured males^{27,28}, compared to areas with low predation. Males exhibit different reproductive behaviours in high versus low predation areas, change from visual to non-visual behaviour (i.e. sneaking copulation)²⁹, and shift conspicuous displays to times of day when light intensity is low³⁰. Furthermore, several life history traits differ between populations from localities differing in predation risk³¹. Guppies from sites where predation intensity is high devote a larger percentage of their body weight to developing offspring (guppies are live bearing), have shorter inter-brood intervals and produce more and smaller offspring compared to fish from low-predation areas.

Female fireflies suffer increased risks of being caught by a predator when signalling for males³². In the species *Photinus collustrans*, females inhabit burrows in the soil and are most vulnerable to predators when courting and copulating outside the burrow. Adaptations to minimize the risk of predation include very brief copulation duration compared to other fireflies, only one copulation instead of repeated mating, and signalling only at the peak of male availability.

Several examples of coevolution between predatory firefly species (*Photuris* spp.) and their prey have been found³³. Female *Photuris* have developed various methods of catching other fireflies, especially males, such as aiming at their luminescence, emitting false signals while approaching signalling males in the air, and mimicking the female signals of the prey species. The prey species, in their turn, have responded evolutionarily to these threats in different ways. Behavioural adaptations in male fireflies to increase the chances of safe mating can be, for example: (1) to shut the light completely when approaching an answering light, (2) to move quickly and fly erratically and unpredictably after flashing, (3) to fly fast while flashing, (4) to land at some distance from the answering light, (5) to test the answering individual by emitting 'nonsense'

Table 2. Adaptations in the guppy *Poecilia reticulata* to decrease predation during reproduction

Trait ^a	Ref.
Rate of courtship	25
Colour patterns and intensity	26
Female preference for brightly coloured males	27, 28
Proportion of visual versus non-visual mating	29
Time of day used for courtship displays	30
Life history traits	31

^a Traits that differ between locations with high and low predation pressure.

(females of their own species would not respond, but the predator would).

It has been suggested that communal sexual displays, such as bird leks and frog choruses, are adaptations that have evolved to decrease the risk of predation during reproduction⁶, but other explanations have also been discussed. In the Guianan cock-of-the-rock (*Rupicola rupicola*), lek size and predation risk were negatively correlated³⁴, which could support the predation hypothesis. This was also the case in the frog *Physalaemus pustulosus*⁶, but since mating probability increased with chorus size, the decreased risk of predation might just be a fortunate consequence rather than a causal factor.

Adaptations that may decrease predation risk during reproductive activities include the ability to change behaviour in the short term, i.e. when predators are actually present. Water striders (*Gerris remigis*) decreased mating duration as well as the number of matings in the presence of predatory fish¹⁹. Habitat switch in response to predation risk was demonstrated in the semi-aquatic bug *Microvelia australina*³⁵. Calling males of the Túngara frog (*Physalaemus pustulosus*) showed a gradient of evasive behaviours in relation to the probability of predation by bats⁶: their response depended on predator distance, bat species and light condition, and ranged from stopping calling to diving underwater and swimming away.

Change in risk taking with age

When a trade-off has to be made between reproduction and predator avoidance, the degree of risk taking during the present reproductive event should be related to the probability of future reproductive opportunities³⁶. In the marine fish *Gobius niger*, young individuals refrained from reproducing when exposed to a predator, while older individuals readily spawned in the same situation³⁶. It was suggested that the higher risk taken by the older gobies depended on their lower probability of future reproduction compared with younger individuals. Furthermore, during one reproductive cycle, the reproductive value of offspring increases as

they mature, due to their increased survival probability. Thus, for animals with parental care, the parent is expected to increase risk taking with time spent with one particular brood³⁷.

↖ *cria*

Future studies

Studies on how animals change their foraging behaviour in the presence of predators are quite numerous (reviewed in Ref. 38), and the importance of predation risk for foraging tactics, locomotory activity, and food and habitat choice is well documented. Also, an increasing number of studies regarding reproductive behaviour show that reproducing animals indeed suffer an increased risk of predation (Table 1). However, only in very few cases has it been shown that individuals actively change their mating behaviour according to predation risk^{6,19,35}. In future, studies of the trade-offs facing reproducing animals should also take into account predator avoidance, since this factor can have a great effect on lifetime reproductive success.

Decision making in reproducing individuals under different circumstances should also be investigated. Dynamic modelling can be used to calculate optimal decisions in situations with conflicting demands. Recently, such a model concerning the trade-off between parental investment and future reproductive success has been developed³⁹. This model deals with parental care in terms of defense and provisioning of offspring in the presence of some kind of risk. However, the same kind of model would be useful also for decisions at an earlier stage, such as during mate searching and copulation. Alternatively, individual differences could be interpreted along the line of Zahavi's handicap principle⁴⁰, with signals for mate attraction being costly in terms of predation risk and revealing a potential partner's quality with respect to ability to avoid predators.

There is a strong taxonomic bias to research on predation risk in reproducing animals. Insects are by far the most-studied organisms, followed by amphibians and fish. In birds, except for lekking species, few studies deal with adult mortality caused by predation during reproduction. In mammals, females in

oestrus have been found to be more vulnerable to detection by predators compared with other individuals¹⁵, but otherwise little is known. Clearly, much remains to be done.

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'Intentional' Signaling in Animal Communication

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The problem of 'intentional' signaling is of interest to evolutionary biologists concerned with animal conflicts and to cognitive ethologists attempting to determine whether nonhuman animals exhibit evidence of complex mental states. Recent theoretical and empirical advances in this area have led to a number of important insights concerning the nature of competitive social interactions. Specifically, and in contrast to earlier claims, individuals involved in social conflicts often produce displays that predict their subsequent behavior, and such predictive displays occur even when repeated social interactions are infrequent. Moreover, studies on a taxonomically diverse set of organisms show that deception, in the form of withholding information, is prevalent. Deception through concealment of information may be widespread because it is more difficult to detect than active falsification of information.

During the 1970s, biologists using the conceptual framework of game theory^{1–3} raised an objection to the claim⁴ that animals should reveal information about their 'intentions' to perform a given behavior. They argued that in intraspecific conflict situations, signals conveying information about the probability of subsequent behavior (one operational definition of intention) would not be favored by selection because they could readily be exaggerated or concealed and thus would become unreliable signals over time. These early models stimulated much work on a wide diversity of species.

The term 'intentional' has also been used to describe a different

set of observations and experiments on animal communication. Research in this area has raised the possibility that the outcome of some social interactions is determined by complex mental processes rather than being the result of individuals responding to predictable situations.

Two senses of intentional signaling

When ethologists⁴ or behavioral ecologists³ discuss an animal's intentions, they typically mean that a given signal is predictive of the signaler's subsequent behavior. For example, Maynard Smith² defined intentions as information about 'the level to which they [signalers] would escalate', or intentions 'specify what an animal will do, or attempt to do, and can be changed with little cost in fitness'⁵. The latter qualification is important, although difficult to measure in practice, as it distinguishes information about intentions from that about fighting ability, or what has been termed resource-holding potential (RHP)⁶. There is little dispute that animals should reliably inform about, and respect, RHP⁷ because most attributes of RHP cannot be altered by animals without incurring significant costs. Instead, controversy has centered on the idea that animals should not provide information regarding subsequent behavior (see Box 1).

The other common usage of the term intentional is derived from philosophy and, as used by some ethologists^{12,13}, raises the possibility that complex mental processes guide the behavior of some animals. A useful framework for investigating intentionality in animals

is Dennett's 'intentional stance'¹⁴, which posits a hierarchy of increasingly complex mental states underlying behavior. Using this framework, the alarm-call system of the vervet monkey (*Cercopithecus aethiops*)^{15,16} has been scrutinized¹³. Observations do not support a higher-order intentional interpretation (e.g. A wants B to believe that A believes there is a leopard). Rather, results suggest that the production of alarm calls is guided by a motivation to inform others about danger or an appropriate escape response (e.g. A wants B to run up into a tree due to the presence of a leopard). As discussed below, however, alarm-calling behavior in vervet monkeys and other nonhuman animals is not simply a reflection of emotional state, since call production appears to be contingent upon the caller's 'audience'. As a result, animals can withhold information about predators and can use alarm calls that are functionally deceptive (see Table 1).

Thus, there are two quite different concepts associated with the use of 'intentional signaling'. To avoid confusion, it would be preferable if researchers were to use *predictive* signaling when discussing the association between a signal and the signaler's subsequent behavior, and *intentional* signaling when discussing the mental processes underlying signal production.

Cases of predictive signaling

In contrast to early predictions from game theory, recent theoretical⁸ and empirical research on agonistic interactions strongly suggests that in species where individuals engage in repeated interactions, signals accurately reveal information about subsequent behavior. However, two studies have also suggested that repeated interactions are not a necessary pre-

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